Water Security 28 (2026) 100213

Contents lists available at ScienceDirect

Water Security

journal homepage: www.sciencedirect.com/journal/water-security

Rainwater harvesting systems for agricultural water security: a systematic review of productivity and resilience

Franco Felix Caldas Silva [a], Fernando Arão Bila Júnior [a], Luís Filipe Sanches Fernandes [a], Fernando António Leal Pacheco [b,*]

[a] Centre for the Research and Technology of Agroenvironmental and Biological Sciences, CITAB, Inov4Agro, Universidadede Trás-os-Montes e Alto Douro, UTAD, Quinta de Prados, 5000-801 Vila Real, Portugal
[b] Center of Chemistry of Vila Real - CQVR, University of Trás-os-Montes e Alto Douro, Ap. 1013, 5001-801, Vila Real, Portugal

ARTICLE INFO

Keywords:
Water scarcity
Decentralized water systems
Climate adaptation
Irrigation efficiency
Dryland agriculture
Rainwater harvesting

ABSTRACT

Water scarcity is intensifying in rainfed and semi-arid agricultural regions, where increasing water demand and climate variability threaten food production and rural livelihoods. In this context, decentralized water storage systems, particularly Rainwater Harvesting (RWH), have emerged as key strategies for agricultural water security. However, the current literature remains fragmented, prioritizing hydrological performance over agricultural, socioeconomic, and long-term system perspectives. This systematic review critically assesses the role of RWH systems in agricultural water security. The review follows the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA 2020) protocol to ensure transparency and reproducibility and applies the Methodi Ordinatio to rank studies. A total of 22 peer-reviewed articles from the Scopus, ScienceDirect, and SciELO databases were analyzed. The results indicate that RWH contribute to increased water availability, and increased agricultural productivity. The RWH systems allow for greater crop intensity and reduce vulnerability to intra-seasonal climate variability. Despite these benefits, significant gaps remain in the literature such as limited socioeconomic integration, a lack of participatory approaches, and an absence of long-term evaluations. However, only 9 studies explicitly evaluated livelihood outcomes, and 11 conducted a limited socioeconomic analysis. The results suggest that the contribution of RWH systems to water security cannot be fully understood solely through technical indicators. Integrated approaches combining hydrological, agronomic, and socioeconomic perspectives are necessary. This study proposes a research agenda focused on interdisciplinary methods, participatory frameworks, and long-term systemic analyses to strengthen agricultural resilience in the face of increasing water scarcity.

1. Introduction

Sustainable water resource management is a 21st-century challenge, with increasing pressure on agriculture due to food demand and growing water scarcity [1,2]. The transition to integrated approaches and the use of decentralized solutions, such as Rainwater Harvesting (RWH), are fundamental to strengthening territorial resilience in times of climate change [3–5].

This systematic review article examines the role of RWH as a disruptive technology for agricultural water security.

1.1. Water security and global water scarcity

Global water scarcity has intensified over the years as a result of increased demand, population growth, urbanization, and climate variability, characterizing itself as a global emergency [6,7]. In regions, particularly those with semi-arid and arid environments, water availability becomes less predictable, leading to structural imbalances between supply and demand. From a water security perspective, scarcity is defined not only by physical limitations but also by the inability to guarantee access to water of adequate quality for human, agricultural, and ecological needs [8,9].

Agriculture remains the largest source of freshwater consumption worldwide, accounting for approximately 70% of total withdrawals.

* Corresponding author.
E-mail addresses: lfilipe@utad.pt (L.F.S. Fernandes), fpacheco@utad.pt (F.A.L. Pacheco).

https://doi.org/10.1016/j.wasec.2026.100213
Received 27 April 2026; Received in revised form 18 July 2026; Accepted 19 July 2026
Available online 29 July 2026
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This leading role in consumption makes agricultural systems particularly vulnerable to hydrological variability and drought conditions [10,11]. As climate change affects precipitation patterns and increases the frequency of extreme weather events, there is a trend toward intensified pressure on water resources, directly impacting food security and rural livelihoods [12].

1.2. Agricultural water security

The concept of water security has evolved from a focus on water availability to a broader perspective encompassing societies' ability to ensure sustainable access to water resources while maintaining economic development, social well-being, and ecosystem integrity [13]. One of the most influential conceptualizations was proposed by Grey and Sadoff [8], who define water security as the availability of water in acceptable quantities and quality to support livelihoods, human well-being, socioeconomic development, ecosystem preservation, and protection against water-related risks. This framework emphasizes that water security goes beyond the availability of physical resources and depends on institutional, environmental, and governance dimensions [14].

Within the agricultural sector, these principles require adaptation to account for the specific characteristics of food production systems. This perspective is consistent with the Food and Agriculture Organization [10], which considers sustainable agricultural water management as a fundamental requirement for achieving food security, rural development, and climate adaptation.

Based on this framework, RWH are interpreted not merely as decentralized water storage technologies, but as strategic components that simultaneously strengthen multiple dimensions of agricultural water security [15,16]. Throughout this review, the contribution of RWH systems is therefore assessed not only in terms of hydrological performance, but also regarding their capacity to improve agricultural productivity, enhance adaptive capacity, support rural livelihoods, and increase the long-term resilience of agricultural systems.

According to Grey and Sadoff [8] and subsequent FAO frameworks, agricultural water security can be understood as a multidimensional concept involving water availability, accessibility, reliability, efficient utilization, and resilience. This review adopts this conceptual perspective as an analytical framework. By examining the current evidence through these interconnected dimensions, the review extends the predominantly technical perspective found in the literature and provides an integrated assessment that links hydrological, agronomic, and socioeconomic aspects of rainwater harvesting systems.

1.3. Agriculture as a key driver of water demand

Agricultural expansion and intensification have significantly increased the demand for water, especially in regions dependent on rainfed systems [7]. As a result, the variability of rainfall patterns generates structural imbalances between the water needs of crops and water availability. These imbalances result in reduced productivity, greater vulnerability to drought, and long-term degradation of natural resources [17].

Conventional irrigation systems, sometimes strictly dependent on groundwater extraction, have contributed to unsustainable water use patterns, including aquifer depletion and deterioration of water quality [18]. As a result, there is an increased need for alternative water management strategies that can increase efficiency while maintaining ecological balance and supporting agricultural productivity [19].

1.4. Role of decentralized water solutions

Decentralized water storage and management systems are a subject of growing interest in proposing viable solutions to address the current challenges posed by increasing water scarcity. Among them, RWH systems stand out for their economically viable and adaptable approach that captures and stores runoff for later use [20,21]. These systems include agricultural ponds, tanks, small reservoirs and other localized infrastructures, designed to increase water availability in the territory [22].

In addition to their hydrological function, RWH systems play a key role in strengthening territorial resilience to climate variability. By providing supplemental irrigation during periods of drought, they help stabilize agricultural production, reduce dependence on groundwater and conventional distribution systems, and improve water management efficiency [23]. Thus, RWH systems are conceptually treated as key components of decentralized water infrastructure in contexts of intensified territorial aridity and water scarcity and in the pursuit of agricultural security [24].

1.5. Research gap and objectives

Despite the increased interest and scientific output on RWH, current studies tend to focus on isolated aspects, such as the suitability of the system's implementation site, hydrological modeling, or infrastructure design. There is a gap regarding integrated analyses that simultaneously consider agricultural productivity, socioeconomic impacts, and governance dimensions.

Critical gaps also persist regarding farmer participation, long-term sustainability, and the role of RWH systems in broader water security frameworks. This fragmentation limits the ability to develop integrated strategies that align technological solutions with applied agricultural and institutional contexts.

In response, this study aims to:

- systematically review the scientific literature on RWH systems in agriculture;
- evaluate their contributions to water security in terms of productivity, efficiency, and resilience;
- identify key thematic trends and methodological approaches;
- Highlight critical gaps and propose a future research agenda

1.6. Scope and limitations of the review

This review focuses exclusively on peer-reviewed journal articles addressing RWH systems in the context of agricultural water security. Conference papers, book chapters, technical reports, and grey literature were not included. Furthermore, only studies indexed in Scopus, ScienceDirect, and SciELO and meeting the predefined eligibility criteria were considered. Consequently, although the review provides a rigorous synthesis of the available studies, relevant papers published outside these databases or in other publication formats may not have been captured. The findings should therefore be interpreted within the scope of the adopted search strategy and selection criteria.

2. Methodology

The overall methodological workflow and study selection process are presented in Fig. 1. The review protocol was prospectively registered in the Open Science Framework (OSF) (DOI: 10.17605/OSF.IO/WMXP5), ensuring transparency and reproducibility of the methodological procedures.

Fig. 2 shows the detailed general approach developed in the work.

The following sections show the methodological strategy used in this study.

2.1. Systematic review design

This study adopts a systematic literature review approach to ensure transparency, reproducibility, and rigor in the selection and analysis of scientific publications. The review is considered systematic because it

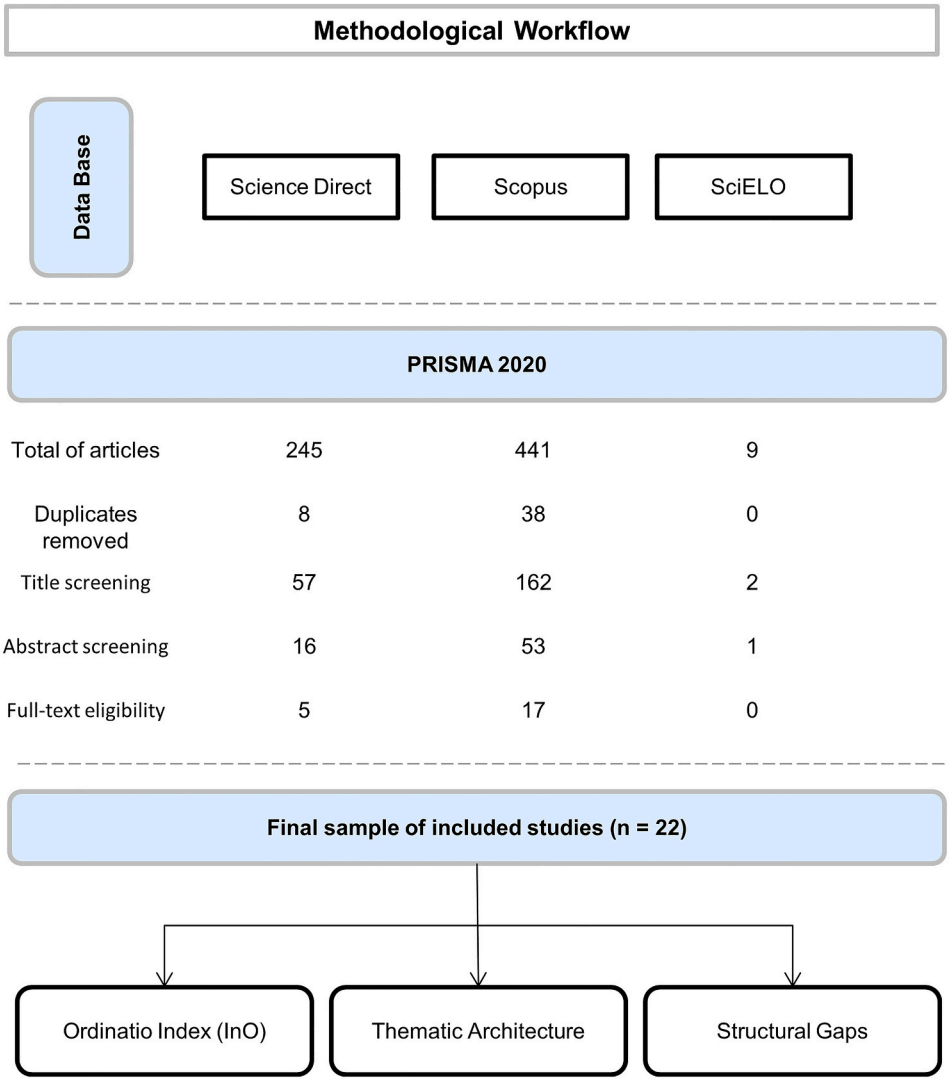


Fig. 1. Integrated methodological workflow combining PRISMA 2020 screening, database consolidation, and analytical framework (Methodi Ordinatio, thematic classification, and gap identification).

follows a predefined review protocol based on the PRISMA 2020 guidelines [25], including explicit search strategies, eligibility criteria, screening procedures, and study selection steps.

The PRISMA 2020 protocol enables the organization of the review process into clearly defined stages, including identification, screening, eligibility assessment, and final inclusion of studies. The study selection process was carried out through independent dual screening, with two reviewers assessing the articles separately. Any discrepancies were resolved through discussion to ensure consistency and methodological rigor.

2.2. Data sources and search strategy

The bibliographic search was conducted in three scientific databases: Scopus, Science Direct, and SciELO. These databases were selected to ensure coverage of peer-reviewed publications in varied geographic contexts.

The search queries combined keywords related to rainwater harvesting, water security, and agricultural systems. The search strategy was designed to capture studies that addressed the technical and socio-environmental dimensions of rainwater harvesting systems.

The search in the 3 databases was performed between March 23 and 27, 2026, and the following keywords and wildcard character were used: “rainwater harvesting”; “runoff harvesting”; “stormwater harvesting”; “retention basin*”; “detention basin*”; “farm pond*”; “check dam*”; “small reservoir*”; “agriculture”; “irrigation”; “rainfed agriculture”.

The Boolean operators “AND” and “OR” were also used.

The review included only peer-reviewed research articles and review articles published between 2021 and 2026. This publication period was selected to capture the most recent advances in Rainwater Harvesting systems for agricultural water security, particularly in the context of climate adaptation and sustainable water management.

2.2.1. Inclusion and exclusion criteria

During the title-screening stage, records were excluded when they clearly did not address RWH or related water harvesting systems in agricultural contexts. This assessment was based on the predefined eligibility criteria rather than solely on the presence or absence of search terms in the title. Titles referring exclusively to urban drainage, drinking water supply, flood control, wastewater management, or unrelated hydrological topics were excluded at this stage. Title screening was independently performed by two reviewers, and disagreements were resolved through discussion until consensus was reached.

Studies were selected based on the following inclusion criteria:

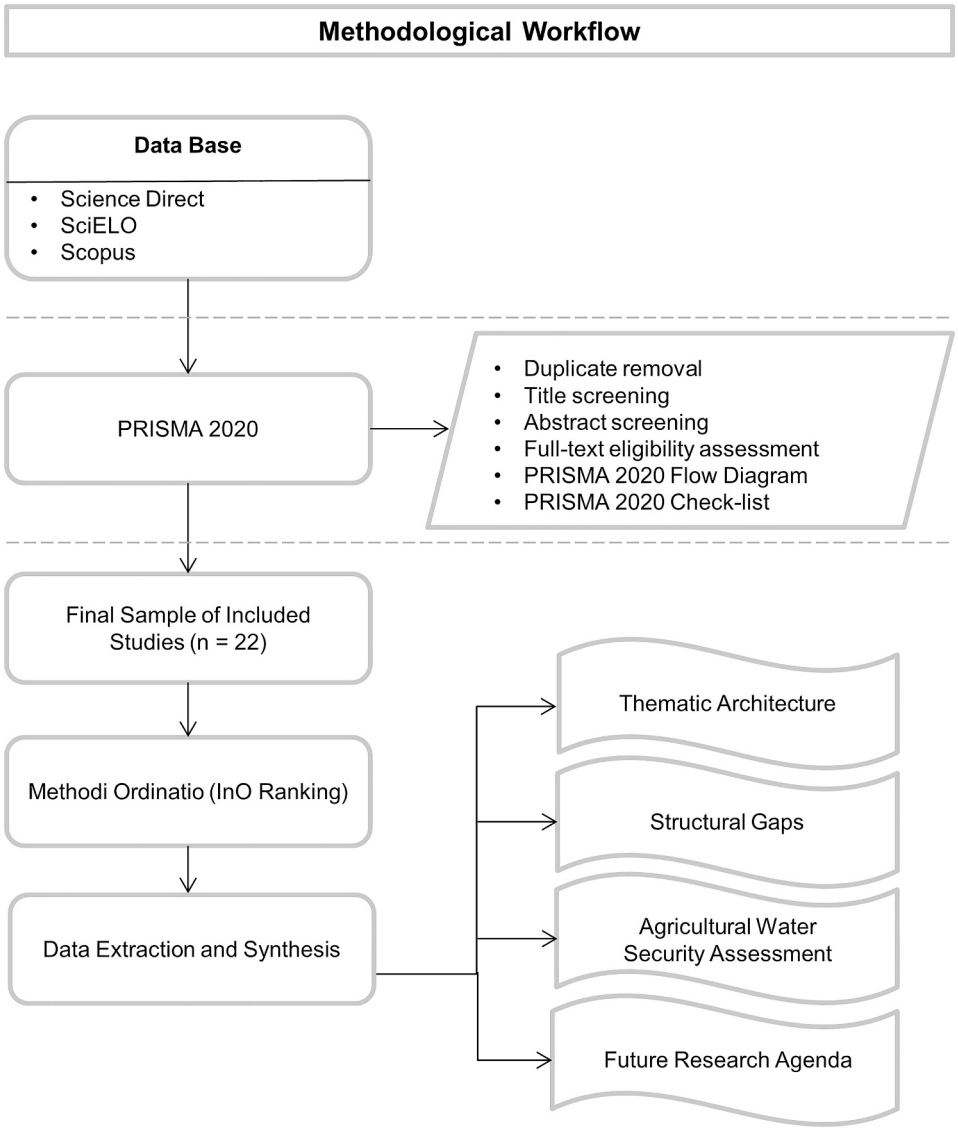


Fig. 2. Methodological workflow adopted in the systematic review. The workflow summarizes the sequential stages of the review.

- focus on rainwater harvesting or decentralized water storage systems;
- application within agricultural contexts;
- assessment of impacts on productivity, water use, or resilience;
- peer-reviewed publications.

Exclusion criteria included:

- studies focused solely on site suitability without agricultural analysis;
- purely hydrological studies lacking application to agricultural systems;
- studies not available in full text.

1.1.1. ***Screening and selection process***

The screening process followed the PRISMA 2020 framework, including duplicate removal, title and abstract screening, and full-text eligibility assessment. The final selection of 22 studies reflects a focused and high-quality body of literature addressing the intersection between rainwater harvesting and agricultural water security. This intentional focus enables a granular analysis of structural gaps that broader reviews may overlook. The PRISMA 2020 checklist used to guide the reporting of this review is provided as supplementary material.

2.2.2. Ordinatio ranking

To ensure the selection of high-quality and relevant studies, the Methodi Ordinatio [26] was applied. This method classifies scientific publications based on a combination of impact factor, number of citations, and year of publication, ultimately generating an Índice Ordinatio (InO) for each selected paper. Table 1 shows the 22 selected articles and their respective InO.

The map shown in Fig. 3 illustrates the global distribution of articles according to the number of studies produced per country.

The selected articles were subsequently characterized according to their themes and limitations.

Fig. 4 correlates the ten highest-ranked studies according to InO with the number of citations and year of publication.

The distribution shows that highly ranked studies are not necessarily the most cited or the most recent, but rather those that combine scientific impact with publication recency, reinforcing the rationale for adopting Methodi Ordinatio as a complementary procedure for prioritizing relevant evidence in the review.

Although the final dataset comprised 22 studies, this reflects the

Table 1
Characterization of the selected studies ($n = 22$) and their ranking based on the Methodi Ordinatio (InO), considering impact factor, number of citations, and year of publication.

Author	Title	InO	Ranking
A. Das; D. Datta; T. Samajdar; R. G. Idapuganti; M. Islam; B. U. Choudhury; K. P. Mohapatra; J. Layek; S. Babu; G. S. Yadav	Livelihood security of small holder farmers in eastern Himalayas, India: Pond based integrated farming system a sustainable approach	65,00	1
L. Rosa; L. He	Global multi-model projections of green water scarcity risks in rainfed agriculture under 1.5 °C and 3 °C warming	57,01	2
M. Negera; Z. A. Dejen; D. Melaku; D. Tegegne; M. E. Adamseged; A. Haileslassie	Agricultural Productivity of Solar Pump and Water Harvesting Irrigation Technologies and Their Impacts on Smallholder Farmers' Income and Food Security: Evidence from Ethiopia	57,00	3
B. Zongo; B. Barbier; A. Diarra; M. Zorom; C. Atewamba; O. S. Combary; S. Ouédraogo; P. Toé; Y. Hamma; T. Dogot	Economic analysis and food security contribution of supplemental irrigation and farm ponds: evidence from northern Burkina Faso	56,01	4
P. Panwar; D. Machiwal; V. Kumari; S. Kumar; P. Dogra; S. Manivannan; P. R. Bhatnagar; J. M. S. Tomar; R. Kaushal; D. Jinger; P. K. Sarkar; L. K. Baishya; N. P. Devi; V. Kakade; G. Singh; N. R. Singh; S. G. Singh; A. Patel; P. S. Renjith; S. Pal; V. K. Bhatt; N. K. Sharma; O. P. S. Khola; S. K. Radhakrishnan; V. Kasthuri Thilagam; P. L. Bhutia; K. Nath; R. Das; D. Daschaudhuri; A. Kumar; G. S. Panwar; S. V. Dwivedi; S. Kumar; B. K. Singh	Sustainable Water Harvesting for Improving Food Security and Livelihoods of Smallholders under Different Climatic Conditions of India	56,00	5
W. Su; B. Li; X. Ouyang; S. Wu; H. Feng; K. H. M. Siddique	Integrated effects of rainwater harvesting and organic fertilization on soil structure and aggregates in a rainfed orchards on the Loess Plateau	50,01	6
J. Koide; S. Yokoyama; S. Hirouchi; C. Hirose; N. Oka; M. Oda; S. Yanagihara	Exploring climate-resilient and risk-efficient cropping strategies using a new pond irrigation system: An experimental study in northern Ghana	49,01	7
D. R. Palsaniya; S. Kumar; M. M. Das; S. K. Rai; T. K. Kumar; M. Chaudhary; K. Chand; A. Ahmed; C. S. Sahay; P. Kumar	Rain water harvesting, agroforestry and goat based intensification for livelihood resilience in drought prone rainfed smallholder farming system: a case for semi-arid tropics	49,00	8
D. R. Palsaniya; S. Kumar; M. M. Das; T. K. Kumar; M. Chaudhary; K. Chand; S. K. Rai; A. Ahmed; C. S. Sahay	Ecosystem services from rain water harvesting, agroforestry and livestock based smallholder rain-fed integrated farming system	48,00	9
M. Sanogo; R. Yonaba; A. Lawane; M. Zorom; F. Tassembédo; H. Ali Sahad; I. Bazié	Do Runoff Water Harvesting Ponds Affect Farmers Cropping Choices? Insights from Smallholders in the West African Sahel	47,00	10
D. Rizzo; F. Vinatier; F. Jacob; I. Ferchichi; I. Mekki; J. Albergel; J.-S. Bailly	A framework for the sustainable maintenance of permanent runoff management structures in rainfed agriculture under climate change	46,01	11
M. Manikandan; B. B. Saliha; B. Narsimlu; J. V. N. S. Prasad; K. Baskar; V. Sanjivkumar; S. Manoharan; G. Guru; G. R. Chary; K. V. Rao; R. Rejani; V. K. Singh	Design of Rainwater Harvesting Pond for Runoff Storage and Utilization in Semi-Arid Vertisols	45,00	12
N. Noor; Z. Wanchang; N. H. Chandio; G. A. Mahar; G. Rahman	Conservation and harvesting of rainwater for sustainable agriculture in Kachho, Sindh, Pakistan	45,00	13
S. F. Demessie; M. Woldeyohannes	Evaluation of water productivity of smallholder irrigation along a Sand River, in the Maigobo watershed of northern Ethiopia	44,00	14
J. B. Nana; H. M. Abd El Baki; H. Fujimaki	Designing an Economical Water Harvesting System Using a Tank with Numerical Simulation Model WASH_2D	41,00	15
A. Deo; S. Karmakar; A. Arora	Rainwater harvesting and water balance simulation-optimization scheme to plan sustainable second crop in small rain-fed systems	40,01	16
S. Halder; S. Das; S. Basu	EFFECTIVENESS OF HAPA AS SMALL RAINWATER HARVESTING PONDS FOR SUSTAINABLE AGRICULTURAL DEVELOPMENT IN DIVERSIFIED ARID REGIONS: COMPARATIVE CASE STUDIES FROM BANKURA IN INDIA	37,00	17
A. Mullo-Sinaluisa; C. Oquendo-Borbor; A. Velastegui-Montoya; B. Merchan-Sanmartín; M. Chávez-Moncayo; V. Herrera-Matamoros; P. Carrión-Mero	Hill Dam Design to Improve Water Use in Rural Areas—Case Study: Sacachún, Santa Elena	36,00	18
S. K. Rautaray	Field Design for Enhancing Water Productivity in Waterlogged Areas with Efficient Water Harvesting and Farming System	33,00	19
D. H. Ranade; M. L. Jadav; I. Swarup; O. P. Girothia; D. V. Bhagat; A. Upadhyay	Crop Productivity Enhancement under Soybean Based Cropping System through Harvested Rain Water in Malwa Region	31,00	20
M. F. Jilito; D. Y. Wedajo; B. W. Feyisa; T. W. Tuke	Water storage practices for small-scale irrigation systems in East Hararghe zone, Ethiopia	30,00	21
M. Kang; C. Yoo; W. Na	Assessment of the contribution of rainwater collection to crop production on udo island, korea	27,00	22

strict eligibility criteria adopted in this review rather than a limited body of literature. Numerous publications on RWH were excluded because they did not explicitly address agricultural water security or lacked empirical evidence relevant to the review objectives.

The combined use of PRISMA 2020 and Methodi Ordinatio enhances both transparency in study selection and robustness in literature

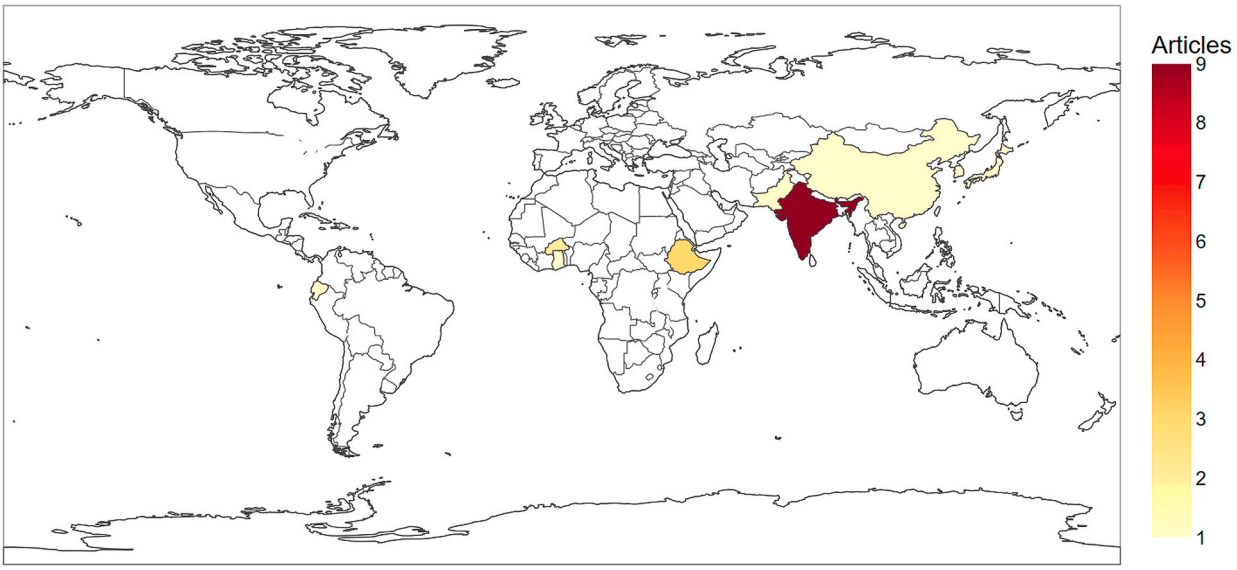


Fig. 3. Geographic distribution of the 22 studies included in the systematic review. Countries are shaded according to the number of empirical studies identified.

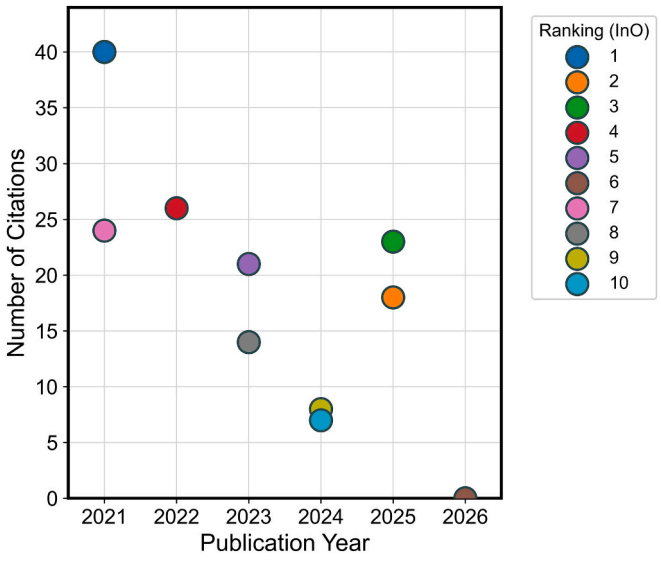


Fig. 4. Distribution of the ten highest-ranked studies according to the Methodi Ordinatio (InOr) index. The figure presents the relationship between publication year and number of citations for the ten highest-ranked studies.

prioritization.

2.2.3. Thematic coding and gap analysis

Following the selection of the final sample, each article was systematically examined to identify its principal research themes and the knowledge gaps explicitly reported by the authors or identified during the systematic analysis of each study. The thematic coding was based on predefined analytical categories established according to the objectives of this review. Four research themes were considered: (i) impacts on agricultural productivity, (ii) models and analysis tools, (iii) scenario projection, and (iv) economic assessment. In parallel, four recurrent knowledge gaps were identified: (i) superficial agricultural approach, (ii) limited socioeconomic analysis, (iii) absence of farmer involvement, and (iv) lack of future projections.

Because individual studies frequently addressed more than one research theme and exhibited more than one knowledge gap, multiple coding was permitted. Each article could contribute more than one occurrence for each identified theme and gap. The thematic matrix was subsequently used to generate the Sankey diagram, in which the width of each flow is proportional to the number of studies sharing a given theme–gap relationship. The complete coding matrix for all 22 reviewed articles is provided as Supplementary Material.

3. Rainwater harvesting systems in agriculture

The 22 selected articles analyze RWH systems not only as isolated reservoirs, but as decentralized resilience infrastructures. These systems operate at the water-food nexus, acting proactively with the goal of using captured water to support agriculture.

A bibliometric mapping of co-occurrence keywords was carried out to contextualize the research landscape and identify the temporal evolution of the area (Fig. 5).

The network visualization reveals a robust and highly interconnected core centered on ‘rainwater harvesting’, ‘environmental science’, and ‘agricultural productivity’, confirming the multidisciplinary nature of the study.

The overlay visualization, which takes into account the average year of publication (from 2021 to 2025), highlights a significant shift in research focus. While fundamental topics such as ‘irrigation’, ‘ecology’, and ‘hydrology’ (represented in dark blue/green) formed the basis of the initial literature, there is a clear emergence of more specialized and urgent themes in the 2024–2025 period (represented in yellow). These recent trends emphasize ‘water content’, ‘soil fertility’, and ‘livelihoods’, signaling a transition from purely technical assessments to the practical application of water security in food systems.

However, the peripheral positioning of nodes such as ‘climate change’, ‘economics’, and ‘mathematics’ suggests a persistent fragmentation in the literature. This spatial distribution provides empirical evidence for the knowledge gaps addressed in this study.

The following sections address in detail the RWH methods applied to agriculture.

3.1. Types of RWH systems

The selected studies reveal diversity regarding the RWH systems adopted and applied in agricultural contexts, reflecting different hydrological conditions, technological levels, cultivation strategies, and local needs. Instead of functioning as isolated storage units, the identified systems operate as decentralized infrastructures that increase water

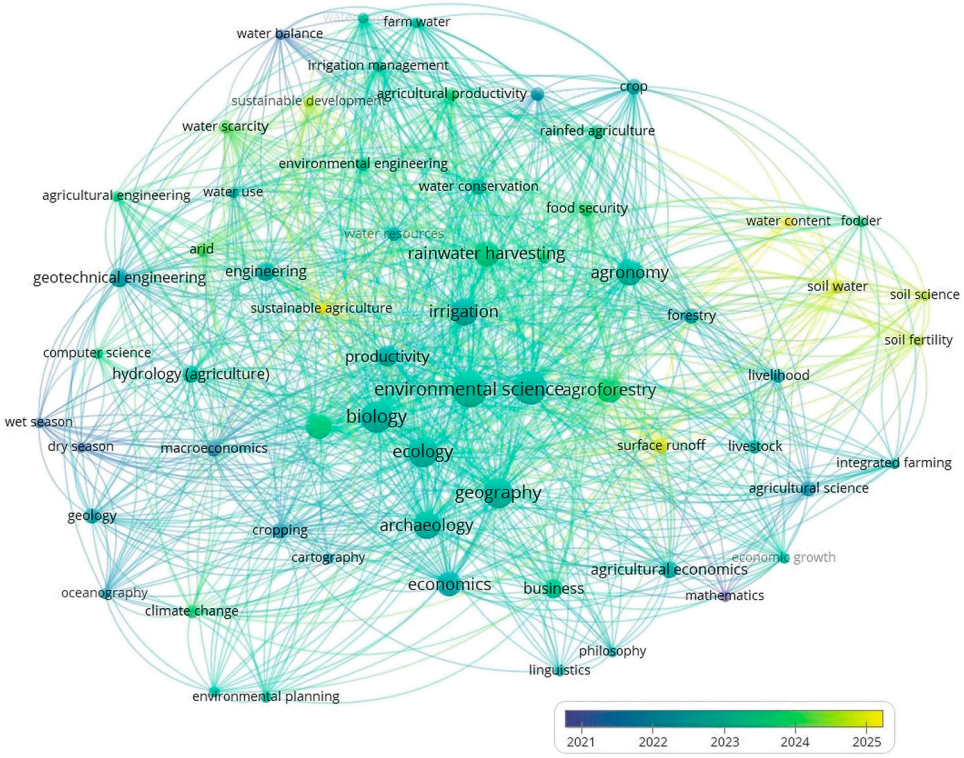


Fig. 5. Overlay visualization map of keyword co-occurrence. The node size represents the frequency of the terms, while the colors indicate the average year of publication.

availability, with the capacity to stabilize agricultural production and support adaptive responses to water scarcity.

Based on their functional characteristics, application, and integration into agricultural systems, RWH technologies can be grouped into five main categories.

The first group comprises on-farm storage systems, which were the most recurrent among the selected studies. These include agricultural tanks, surface runoff catchment ponds, small-scale reservoirs, and localized storage systems such as hapa ponds. These structures are typically designed to capture and store surface runoff for supplemental irrigation, particularly during periods of drought. Their relevance lies in their simplicity, relatively low cost, and direct impact on water availability at the field scale [27–36].

The second category includes integrated water production systems, in which RWH structures are embedded within broader agricultural systems. Examples include Integrated Farming Systems (IFS), agroforestry systems, and crop-livestock integration models based on ponds. In these cases, water harvesting is not an isolated intervention but is embedded in a multifunctional system that simultaneously increases productivity, diversifies income sources, and improves ecosystem services [37–41].

The third group addresses hydro-ecological and landscape systems, such as sand dams, slope dams, and water management interventions in wetlands. These systems operate on a broader spatial scale and are designed to modify hydrological processes, including aquifer recharge, surface runoff regulation, and soil moisture dynamics. Their contribution extends beyond individual rural properties and influences water availability at the watershed level [42–45].

The fourth category encompasses hybrid and multi-source water systems, with the interaction between different water harvesting and storage techniques, including tanks, ponds, shallow wells, and artesian boreholes. These systems reflect a strategy of diversifying water sources, aiming to reduce vulnerability to rainfall variability and increase system reliability under uncertain climatic conditions [46,47].

The fifth category deals with climate impact assessment and regulatory frameworks, being an analytical and strategic group. These studies use predictive modeling and risk analysis to quantify future water availability (such as 'green water') and the vulnerability of global food security, providing a basis for public policies and intervention priorities in the face of climate change [48].

It can be observed that the diversity of rainwater harvesting systems identified in the selected studies highlights that there is no single ideal solution, requiring a socio-economic-environmental analysis of the area under analysis. The system's effectiveness will depend on the alignment between the technological design, environmental conditions, and agricultural practices. This variability reinforces the need for specific approaches for each context, while also connecting them on a larger scale.

3.2. *Application contexts and geographic distribution*

Analysis of the selected studies reveals that RWH systems are implemented in a wide variety of geographic contexts, mainly in arid and semi-arid regions of the Global South. The geographic distribution shows a high concentration of research developed in Asia and Africa, which reflects the structural need for decentralized water solutions in areas where rainfed agriculture is notoriously relevant.

India is the country with the largest number of studies [30,32,33,36,38,40–42,46], with interventions applied in several states, such as Uttar Pradesh, Maharashtra and Assam. Other significant geographic contexts include China [39], Pakistan [35] and South Korea [34] in Asia; Burkina Faso [28,29], Ghana [37] and Ethiopia [27,44,47] in Africa; and Ecuador [45] in South America.

Regarding the scale of intervention, studies demonstrate a dual approach:

- Microscale and Individual Plots: Many systems are designed for high-precision impact at the farm level, often using standardized 1-ha areas [32,38,40] or specific storage capacities, such as "mool-

tongs" in South Korea [34]. Smaller experimental scales, such as 72 m^2 plots in Japan [31] or 1000 m^2 plots in India [33], focus on evaluating the ideal reservoir volume and the cost-benefit ratio.

- Community and Regional Scale: Several studies analyze rainwater harvesting as a collective infrastructure, involving groups ranging from 12 to 22 producers [28,36] or entire communities [27,30,35,37,41,42,44,45].

This highlights the social dimension of rainwater harvesting, where the management of harvested water requires community governance and shared benefits.

The observed geographic and scale diversity reinforces the role of RWH as a versatile technology, adaptable to different socio-ecological landscapes, from individual smallholder plots to landscape-wide community interventions.

4. Impacts on agricultural water security

The integration of RWH into agricultural systems acts as a catalyst for water security, transforming precipitation into a manageable resource. The selected studies indicate that the transition from exclusive dependence on rainfall to managed water storage mitigates "Green Water Scarcity" (GWS). The following sections address water and agricultural efficiency in the analyzed scenarios.

4.1. Effects on productivity and water use efficiency

The systematic review of selected studies reveals that RWH is a key factor in increasing "production per drop" in smallholder farming systems. The transition from rainfall dependence to managed supplemental irrigation has resulted in substantial productivity gains in different agroclimatic zones. In India [41], pond-based Integrated Farming Systems (IFS) led to a significant increase in Rice Equivalent Yield (REY), rising from 0,61 to to 4.04 Mg per unit. Similarly, in Pakistan [35], wheat productivity increased from 450 to 815 kg over a seven-year period since the implementation of small water reservoirs.

The data also highlight a significant improvement in Water Use Efficiency (WUE) when integrating RWH with precision technologies. In Ethiopia [27], this combination resulted in a Net Water Value (NWV) of up to 1.53 USD/m^3. In another study conducted in India [32], the combination of Moisture Conservation Practices (MCP), RWH, and micro-irrigation increased system productivity by 25% with a favorable cost-benefit ratio over 13 years.

It is also observed that RWH interventions improved soil water dynamics. For example, in the Loess Plateau [39], the combined effect of RWH with organic fertilization promoted synergistic honey from rainwater harvesting, and organic fertilization improved soil aggregate stability, reduced erosion risks, and increased soil water retention. In Udo, South Korea, the use of mool-tongs reduced the occurrence of water deficit from 30 to 20%.

The studies show the effectiveness and feasibility of implementing RWH at various scales and environments, and report possibilities for integration with other agricultural production techniques.

4.2. Cropping intensity and system resilience

Rainwater management tools demonstrate sustainable intensification and resilience of livelihoods, enabling farmers to overcome rainfall seasonality. The availability of stored water has enabled the diversification from single-crop to multiple-crop systems, with studies reporting crop intensity exceeding 200% [38,46]. This intensification is often characterized by the introduction of high-value dry-season crops, such as vegetables, garlic, and onions, which were previously unviable under rainfed conditions.

System resilience increases through the integration of livestock and agroforestry. The inclusion of goats in integrated agroforestry system models [38] increased net returns by up to 226% in intensive and semi-intensive farming systems.

Economically, the interventions analyzed showed high viability, with cost-benefit ratios reaching 3.38 [46] and payback periods between 6.8 and 12 years [33,37]. Rainwater management allows producers to avoid total crop loss during critical drought periods [32], and rainwater harvesting infrastructure ensures long-term food and income stability for small farmers in areas of greater climate vulnerability.

5. Modeling and analytical approaches

In this field of study, there is a clear segmentation with regard to the methodological approaches adopted.

It is observed that the study that developed a climate projection with different temperature scenarios and the influence of this factor on food production worked at a global level, acting as an initial screening tool that shows the areas with the greatest vulnerability to reduced food production [48].

It is noted that studies that incorporate spatial modeling into their methodology tend not to conduct analyses of agricultural productivity. This behavior can be observed in studies conducted in Ecuador [45], Japan [31], and South Korea [34], where simulations were performed to optimize the RWH structure and water efficiency was analyzed.

Studies conducted in India and Pakistan stand out positively due to their more robust analysis, where, in addition to field studies, hydrological modeling, system efficiency, cost analysis, and productivity analysis were performed [32,35].

On the other hand, there is a group of studies that focuses on the productivity and efficiency of the system. These studies quantify productivity gains and perform cost-benefit analyses, but they do not carry out hydrological modeling, presenting an approach restricted to a socio-environmental reality [27,36,38–41,44].

There is also another approach focused on questionnaires. This approach allows for a robust socio-economic analysis and serves as a tool for characterizing the groups concerned. However, the absence of field analysis becomes a limitation for these methodologies [29,47].

This section shows the fragmentation in the field of study with a very restricted methodological focus.

6. Socioeconomic dimensions of RWH

Despite the growing number of publications on RWH applied to agricultural production, the integration of socioeconomic dimensions remains limited in the selected studies. Among the 22 articles analyzed, only a small portion presents robust and explicit assessments of socioeconomic outcomes, highlighting a critical imbalance in the current research landscape.

The study conducted in Ethiopia [47] offers a comprehensive evaluation of water storage interventions in smallholder farming systems, emphasizing their role in improving access to irrigation, reducing vulnerability to rainfall variability, and supporting livelihood stability. The results indicate that water storage practices contribute not only to increased agricultural production but also to greater income security and reduced exposure to environmental risks. Importantly, the study highlights that the effectiveness of RWH systems depends on local institutional arrangements, access to financial resources, and the farmers' capacity to manage water infrastructure.

Similarly, the study conducted in Burkina Faso [29] provides a detailed analysis of the socioeconomic impacts of decentralized irrigation systems, including rainwater harvesting ponds. The results show statistically significant improvements in household income, food security indicators, and overall well-being among smallholder farmers who adopt these technologies. The study further emphasizes that the economic viability of these systems is closely linked to complementary factors, including access to markets, agricultural inputs, and technical support.

On the other hand, the other studies analyzed focus on biophysical and technical performance indicators (such as water storage capacity, hydrological balance, and agricultural productivity) while treating socioeconomic variables as secondary or implicit outcomes. Although all reviewed studies reported hydrological outcomes and most documented improvements in agricultural performance, the assessment of livelihood outcomes was less consistent across the literature. To illustrate this pattern, Table 2 summarizes the hydrological, agronomic, and livelihood indicators evaluated in each reviewed study.

As summarized in Table X, all reviewed studies evaluated hydrological outcomes, whereas only 9 of the 22 studies (40.9%) explicitly assessed livelihood outcomes using indicators such as household income, food security, or employment. Although improvements in agricultural productivity are widely reported, their translation into broader livelihood improvements is not consistently examined. This imbalance suggests that productivity gains are frequently used as proxies for livelihood enhancement, highlighting the need for more integrated socioeconomic assessments in future research.

Table 2
Comparison of hydrological, agronomic, and livelihood indicators evaluated in the reviewed studies.

Reference	Agronomic indicators	Livelihood indicators	Hydrological outcomes evaluated	Livelihood outcomes evaluated
[43]	Food production	Not evaluated	✓	×
[41]	Food production	Income	✓	✓
[48]	Food production	Not evaluated	✓	×
[39]	Food production	Not evaluated	✓	×
[44]	Biomass production	Not evaluated	✓	×
[36]	Food production	Not evaluated	✓	×
[35]	Food production	Not evaluated	✓	×
[27]	Food production	Benefit–cost ratio; Food security	✓	✓
[42]	Water saving	Not evaluated	✓	×
[32]	Food production	Not evaluated	✓	×
[46]	Crop diversification	Income	✓	✓
[47]	Water saving	Not evaluated	✓	×
[29]	Food production; Crop diversification	Not evaluated	✓	×
[40]	Food production	Income; Employment	✓	✓
[45]	Food production; Crop diversification	Not evaluated	✓	×
[37]	Food production	Income	✓	✓
[30]	Food production; Crop diversification	Employment; Income	✓	✓
[33]	Food production	Not evaluated	✓	×
[28]	Food production	Income	✓	✓
[38]	Food production	Income	✓	✓
[31]	Food production	Income	✓	✓
[34]	Food production	Not evaluated	✓	×

7. Identified gaps in the literature

A systematic analysis of the selected studies reveals pertinent limitations in the discussion that restrict a comprehensive understanding of RWH implementation in an agricultural context. These gaps are not isolated, but reflect structural trends in the literature ($n = 22$).

Systematic mapping of the literature reveals a clear interconnection between established research groups and persistent scientific gaps in the field of RWH applied to agriculture. Fig. 6 illustrates these relationships through a Sankey diagram, highlighting how current studies are distributed across four main themes and four critical limitations.

The impacts on agricultural productivity represent the most robust flow in recent literature, and are also the area where structural gaps are most evident.

The diagram also highlights a significant disconnect in economic assessments. While vital for decentralized systems, this theme is often associated with limited socioeconomic analysis.

7.1. Superficial agricultural analysis

Regarding agricultural analyses, there is a limited evaluation of the results directly associated with RWH. Although studies report increased water availability or storage capacity, few provide detailed assessments of how this additional water translates into sustainable improvements in agricultural productivity [39].

In other cases, agricultural performance is inferred indirectly through indicators such as biomass production [44], without a robust analysis of the variability in crop productivity, cropping systems, or long-term productivity trends. Furthermore, key dimensions such as crop diversification, soil health, and agronomic management practices are poorly explored.

This limitation results in an incomplete analysis of the real contribution of RWH to food production and agricultural sustainability. Without a deep integration of agronomic variables, the effectiveness of these systems in increasing water security at the farm level remains only partially demonstrated.

7.2. Limited socioeconomic integration

The analysis of the selected articles also reveals a clear imbalance between technical and socioeconomic assessments. As discussed in Section 6, only a small portion of the studies explicitly examine the impacts of RWH implementation on farmers' income, food security, and livelihoods [27,40].

The predominant studies implicitly assume that improvements in water availability and agricultural productivity will lead to positive socioeconomic outcomes [45]. However, this assumption is not validated by empirical data. Critical aspects such as producer adherence, cost-benefit, financial accessibility, labor needs, and market integration are frequently neglected.

Furthermore, issues related to equity, including differentiated access to water infrastructure among farmers, are largely absent from the literature. This gap is particularly relevant in smallholder farming contexts, where resource constraints and institutional barriers significantly influence technology adoption.

As a result, the current body of research does not fully capture the human dimension of water security, limiting its applicability to policy-making and real-world implementation.

7.3. Lack of farmer participation

Another relevant gap identified in the literature is the limited incorporation of participatory approaches in the design, implementation, and evaluation of RWH systems. Studies generally adopt top-down

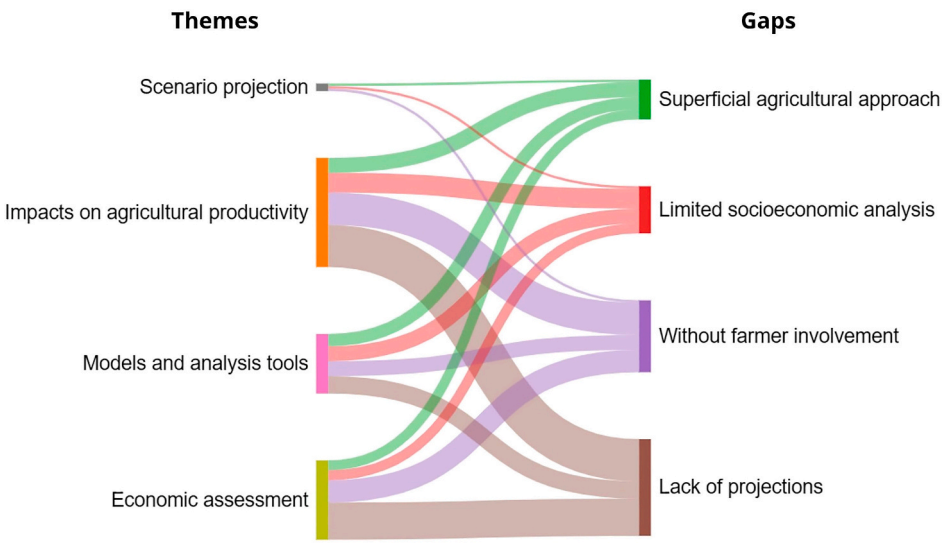


Fig. 6. Sankey diagram illustrating the connectivity between the main research themes (left) and the identified knowledge gaps (right) in the reviewed literature. Each article could be assigned to multiple themes and multiple gaps according to the predefined coding framework. Flow widths are proportional to the number of studies sharing each theme–gap relationship across the 22 analyzed articles. The complete coding matrix is provided in the Supplementary Material.

methodologies, relying on modeling, experimental setups, or externally designed interventions, with few exceptions [29,47].

The knowledge, perceptions, and adaptive strategies of farmers are rarely integrated into the analytical framework. This omission reduces the ability to understand how local decision-making processes influence the adoption, management, and effectiveness of decentralized rainwater management systems.

The lack of participatory research limits the applicability and relevance of interventions, since systems not developed in conjunction with users may face barriers to adoption.

7.4. Absence of long-term projections

A further limitation in the analyzed literature is the scarcity of long-term assessments under changing climatic and socio-environmental conditions. Most studies are based on short-term observations, case studies, or simulations that do not capture interannual variability or the long-term dynamics of the system.

This gap is particularly critical given the increasing uncertainty associated with climate change, including alterations in precipitation patterns, frequency of extreme events, and prolonged periods of drought. Without long-term projections, it becomes difficult to assess the resilience and sustainability of systems over time.

Furthermore, there is a lack of methodologies that integrate climate scenarios, hydrological variability, and future water demand in their analyses. The absence of such prospective approaches limits the ability to assess whether current RWH strategies are robust enough to ensure future water security.

Filling this gap requires the integration of climate modeling, long-term field data, and adaptive management frameworks that consider both environmental and socio-economic dynamics.

8. Implications for water security

The results of this systematic review suggest that RWH systems play a significant role in addressing key dimensions of agricultural water security, especially in areas exposed to water scarcity. However, their contribution goes beyond simple water storage, encompassing broader interactions between hydrological processes, agricultural productivity, and socio-institutional dynamics.

The gaps identified highlight that the current understanding of RWH systems remains fragmented, limiting their full potential as a strategic tool for agricultural water security. Therefore, a more integrated perspective is needed to assess how these systems contribute to resilience, sustainability, and adaptive capacity in the face of increasing climate variability.

8.1. RWH as a tool for climate adaptation

Climate change is increasing the frequency and intensity of droughts, altering rainfall patterns, and exacerbating water scarcity in rainfed and semi-arid agricultural regions. These changes not only increase the demand for decentralized water storage systems but also influence the performance of RWH systems themselves. Changes in rainfall seasonality, greater variability in precipitation, and the increasing occurrence of extreme rainfall events may reduce the predictability of runoff generation, affect storage reliability, and challenge the design and operation of RWH infrastructure. Consequently, the effectiveness of these systems under future climatic conditions depends on their capacity to accommodate increasing hydroclimatic variability. In this context, decentralized RWH systems have gained increasing attention as climate adaptation strategies because they enhance local water storage, reduce dependence on irregular rainfall, and improve the capacity of farming systems to cope with climatic variability.

The reviewed studies generally indicate that RWH systems increase farmers' ability to cope with intra-seasonal variability, reduce crop loss risk, and stabilize agricultural production. However, the magnitude of these benefits varies across agroecological zones. Differences in rainfall regimes, soil characteristics, topography, cropping systems, and local management practices influence the hydrological performance and agricultural outcomes of RWH systems. Consequently, although the reviewed studies generally support the positive contribution of RWH to agricultural water security, the effectiveness of these systems is context-dependent and should be interpreted considering local environmental and socioeconomic conditions. This function is particularly relevant in regions experiencing increasing frequency and intensity of drought events.

However, the effectiveness of RWH as an adaptation strategy depends on its integration with flexible agricultural practices. Static irrigation demands and fixed cropping systems can reduce adaptive capacity, especially under highly variable climatic conditions. Therefore, adaptive water allocation, crop diversification, and climate-responsive management strategies are essential to maximize the resilience benefits of RWH systems.

8.2. Contribution to sustainable water management

Beyond their crucial role in climate change scenarios, RWH contribute to more sustainable water management by promoting the decentralized use of local water resources. These systems reduce pressure on overexploited aquifers and increase water availability at the farm level.

The literature indicates that RWH systems can improve water use efficiency and support increased crop intensity, particularly when combined with efficient irrigation practices. Furthermore, nature-based storage solutions, such as agricultural tanks and small reservoirs, can contribute to aquifer recharge and ecosystem regulation.

However, the sustainability of these systems depends on proper design, operation, and integration with broader water management frameworks. In some cases, increased water availability can lead to intensification patterns that exceed the limits of local resources, potentially resulting in unsustainable water use.

This highlights the need for systemic approaches that consider the interactions between water availability, agricultural demand, and environmental constraints. Sustainable water management through RWH requires not only technical optimization and popular adoption, but also governance mechanisms that regulate water use and ensure the long-term balance of resources.

8.3. Policy and planning implications

The conclusions of this systematic review have direct implications for water policies and agricultural planning. Rainwater harvesting systems represent scalable and economically viable solutions for increasing water security, especially in regions where centralized infrastructure is limited or economically unfeasible.

However, the implementation and expansion of these systems require supportive institutional structures. Important policy considerations include financial incentives for adoption, technical assistance, and the integration of rainwater harvesting into broader water resource management strategies.

Furthermore, the lack of socioeconomic and participatory dimensions in current research suggests that policies should prioritize inclusive approaches that consider farmers' needs, local knowledge, and social equity. Without these considerations, the adoption of rainwater harvesting systems may be uneven, reinforcing existing inequalities in access to water resources.

Effective planning should also incorporate spatial and climatic variability, ensuring that rainwater harvesting interventions are adapted to local conditions. The integration of geospatial tools, climate projections, and participatory planning processes can enhance decision-making and improve the long-term effectiveness of rainwater harvesting strategies.

9. Future research agenda

The research priorities identified in this review are broadly consistent with those reported in the recent global synthesis of RWH research by Chen, et al. [49]. However, whereas that study provides a comprehensive assessment of RWH research across multiple application domains, the present review focuses exclusively on the role of RWH systems in agricultural water security. Within this specific context, our findings confirm several previously identified challenges, including fragmented methodological approaches, weak socioeconomic integration, limited long-term assessments, and the need for stronger governance and institutional support to facilitate the adoption and scaling of RWH systems. At the same time, this review extends previous syntheses by demonstrating that, although improvements in water availability and agricultural productivity are frequently reported, empirical evidence linking these outcomes to farmers' livelihoods and agricultural resilience remains limited. Furthermore, the present review highlights the need for an integrated analytical framework that simultaneously considers hydrological performance, agricultural productivity, farmer participation, and resilience within the context of agricultural water security.

Based on the gaps identified in this review, a structured research agenda is proposed to promote the understanding and application of RWH systems in the context of agricultural water security.

First, future studies should prioritize integrated assessments that combine hydrological, agronomic, and socioeconomic dimensions. The current fragmentation of the literature limits the ability to fully evaluate the performance and impacts of RWH systems.

Second, there is a need to expand the use of participatory and transdisciplinary approaches, incorporating farmers' knowledge and local decision-making processes. These approaches can enhance the planning, adoption, and sustainability of RWH implementations.

Third, long-term and climate-based analyses should be strengthened. This includes integrating climate change scenarios, interannual variability, and system dynamics to assess the resilience of RWH systems over time.

Fourth, future research should focus on governance and institutional frameworks, particularly with regard to access, equity, and regulation of water use. Understanding how policies and institutions shape the adoption and performance of RWH systems is essential for scaling up these solutions appropriately.

Finally, the development of decision support tools that integrate spatial analysis, optimization models, and socio-environmental indicators can improve planning and implementation processes, bridging the gap between research and practice.

10. Conclusions

This study provides a systematic and structured synthesis of the current scientific literature on rainwater harvesting systems in agricultural contexts.

The results indicate that RWH systems have significant potential to increase agricultural productivity, improve water use efficiency, and enhance resilience in water-scarce environments. However, this potential is not fully exploited due to persistent gaps in the literature, particularly related to the superficial treatment of agricultural outcomes, limited socioeconomic integration, lack of participatory approaches, and absence of long-term analyses.

These limitations reflect a broader tendency to evaluate RWH systems primarily from technical and hydrological perspectives, without fully considering their role within complex socio-environmental systems.

By identifying these gaps, this study contributes to the conceptual and analytical advancement of agricultural water security, emphasizing the need for integrated and interdisciplinary approaches. The proposed research agenda highlights essential priorities for future investigations, aiming to support the development of more effective and equitable water management strategies.

It is important to acknowledge that the findings of this review are constrained by the limitations of the available evidence, including the heterogeneity of study designs and the limited integration of socioeconomic variables, as well as by the review process itself. Although the review followed the PRISMA 2020 framework and a structured selection protocol supported by the Methodi Ordinatio approach, the evidence synthesis may still be influenced by database selection, search strategy, publication bias, and the predefined eligibility criteria. Furthermore, because this study is based on a qualitative systematic review rather than a quantitative meta-analysis, no formal sensitivity analysis was performed. Consequently, the robustness of the findings should be interpreted within the scope of the adopted methodological framework.

Ultimately, strengthening the role of RWH in agricultural water security requires not only technological innovation but also the alignment of environmental, social, and institutional dimensions. This integration is essential to address the growing challenges of water scarcity and to

ensure sustainable and resilient food production systems in a context of climate change.

Disclosure of competing interest

The authors declare no conflicts of interest.

CRediT authorship contribution statement

Franco Felix Caldas Silva: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Fernando Arão Bila Júnior:** Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Conceptualization. **Luís Filipe Sanches Fernandes:** Validation, Supervision, Project administration, Methodology, Funding acquisition, Conceptualization. **Fernando António Leal Pacheco:** Visualization, Validation, Supervision, Project administration, Methodology, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Funding

This work is supported by the project LivingSoiLL-Healthy Soil to Permanent Crops LLs. This project has received funding from the European Union's Horizon Europe research and innovation programme under grant agreement No. 101157502.For the authors affiliated with CITAB, this work was further supported by National Funds of FCT–Portuguese Foundation for Science and Technology, under the project UIDB/04033/2025 (DOI:10.54499/UIDB/04033/2025). These authors are also affiliated with Inov4Agro–Institute for Innovation, Capacity Building and Sustainability of Agri-food Production, funded under the project LA/P/0126/2020 (DOI:10.54499/LA/P/0126/2025). Inov4Agro is an Associate Laboratory composed of two R&D units (CITAB & Green U Porto). For the author integrated in the CQVR, the research was supported by National Funds of FCT—Portuguese Foundation for Science and Technology, under the projects UIDB/00616/2025 (DOI:10.54499/UIDB/00616/2025) and UIDP/00616/2025 (DOI:10.54499/UIDP/00616/2025). Additionally, the author Franco Felix Caldas Silva is supported by a PhD Fellowship from FCT–Portuguese Foundation for Science and Technology, with the reference 2025.00758.BD.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.wasec.2026.100213.

Data availability

The data that support the findings of this study are available from the corresponding author upon reasonable request.

References

[1] FAO, The State of Food and Agriculture 2020: Overcoming water challenges in agriculture, 2020.
[2] UN, The United Nations world water development report 2023: partnerships and cooperation for Water, 2023, p. 212.
[3] M.S. Waqas, H.K. Bayabil, N.S. Hailegnaw, S. Hussain, A. Tariq, S. Abubakar, Drought mitigation and livelihood improvement options through rainwater harvesting structures in a rainfed agricultural system, Agric. Syst. 230 (2025), https://doi.org/10.1016/j.agsy.2025.104469.
[4] Q. Yuan, Y. Liu, Y. Qie, C. Hu, Y. Meng, F. Luan, Unlocking global rainwater harvesting potential for safe drinking water access, Nat. Commun. 16 (2025) 11320, https://doi.org/10.1038/s41467-025-66429-w.
[5] J. Das, M.K. Goyal, N. Umamahesh, Water Harvesting, Climate Change, and Variability, in: Handbook of Water Harvesting and Conservation: Basic Concepts and Fundamentals, 2021, pp. 427–446.
[6] UNESCO, The United Nations World Water Development Report 2024: Water for prosperity and peace, 2024.
[7] J. Rockstrom, J. Gupta, D. Qin, S.J. Lade, J.F. Abrams, L.S. Andersen, D. I. Armstrong McKay, X. Bai, G. Bala, S.E. Bunn, et al., Safe and just earth system boundaries, Nature 619 (2023) 102–111, https://doi.org/10.1038/s41586-023-06083-8.
[8] D. Grey, C.W. Sadoff, Sink or swim? Water security for growth and development, Water Policy 9 (2007) 545–571, https://doi.org/10.2166/wp.2007.021.
[9] D. Molden, Water for food water for life: A comprehensive assessment of water management in agriculture, Earthscan and International Water Management Institute (IWMI), London, 2007.
[10] FAO, in: FAO (Ed.), The State of Food and Agriculture 2023, 2023.
[11] M. Hernández Lagana, P. Mejias Moreno, AQUASTAT water data snapshot 2025, Food & Agriculture Org., 2025.
[12] (IPCC), I.P.o.C.C, Climate Change 2022 – Impacts, Adaptation and Vulnerability, Cambridge University Press, 2023.
[13] C. Cook, K. Bakker, Water security: debating an emerging paradigm, Glob. Environ. Chang. 22 (2012) 94–102.
[14] D. Garrick, J.W. Hall, Water security and society: risks, metrics, and pathways, Annu. Rev. Environ. Resour. 39 (2014) 611–639.
[15] E. Bresci, G. Castelli, Water Harvesting in Farmlands, in: Handbook of Water Harvesting and Conservation: Basic Concepts and Fundamentals, 2021, pp. 87–100.
[16] A.M. Al-Omran, A. Al-Khasha, S. Eslamian, Irrigation Water Conservation in Saudi Arabia, in: Handbook of Water Harvesting and Conservation: Case Studies and Application Examples, 2021, pp. 373–383.
[17] (IPCC), I.P.o.C.C, Climate change and land: An IPCC special report on climate change, desertification, land degradation, sustainable land management, food security, and greenhouse gas fluxes in terrestrial ecosystems, 2019.
[18] J.S. Famiglietti, G. Ferguson, The hidden crisis beneath our feet, Science 372 (2021) 344–345, https://doi.org/10.1126/science.abh2867.
[19] FAO, The State of the World's Land and Water Resources for Food and Agriculture ; 978–92–5-140285-6, FAO, Rome, Italy, 2025, p. 148.
[20] D.N. Pandey, A.K. Gupta, D.M. Anderson, Rainwater harvesting as an adaptation to climate change, Curr. Sci. 85 (2003) 46–59.
[21] F. Abdulla, C. Abdulla, S. Eslamian, Concept and technology of rainwater harvesting, in: Handbook of water harvesting and conservation: basic concepts and fundamentals, 2021, pp. 1–16.
[22] S. Chen, H. Sun, X. Chen, Towards a new era of rainwater utilization: implementation status, barriers, and prospects of decentralized rainwater harvesting systems in China, Environ. Rev. 33 (2025) 1–21, https://doi.org/10.1139/er-2024-0114.
[23] J. Rockström, J. Barron, P. Fox, Rainwater management for increased productivity among small-holder farmers in drought prone environments, Physics and Chemistry of the Earth, Parts a/B/C 27 (2002) 949–959, https://doi.org/10.1016/S1474-7065(02)00098-0.
[24] J.O. Alao, S.U. Eze, G.I. Onyenweife, A.A. Ibe, O.J. Otorkpa, D.A. Ayejoto, F. Abubakar, M. Abdulsalami, D.O. Abdulmalik, Enhancing water security through integrated storage mechanisms and rainwater harvesting for sustainable development, Discov. Sustain. 6 (2025), https://doi.org/10.1007/s43621-025-01914-2.
[25] M.J. Page, J.E. McKenzie, P.M. Bossuyt, I. Boutron, T.C. Hoffmann, C.D. Mulrow, L. Shamseer, J.M. Tetzlaff, E.A. Akl, S.E. Brennan, et al., The PRISMA 2020 statement: an updated guideline for reporting systematic reviews, BMJ 372 (2021), https://doi.org/10.1136/bmj.n71.
[26] R.N. Pagani, J.L. Kovaleski, L.M. Resende, Methodi Ordinatio: a proposed methodology to select and rank relevant scientific papers encompassing the impact factor, number of citation, and year of publication, Scientometrics 105 (2015) 2109–2135, https://doi.org/10.1007/s11192-015-1744-x.
[27] M. Negera, Z.A. Dejen, D. Melaku, D. Tegegne, M.E. Adamseged, A. Haileslassie, Agricultural productivity of solar pump and water harvesting irrigation technologies and their impacts on smallholder farmers' income and food security: evidence from Ethiopia, Sustainability 17 (2025), https://doi.org/10.3390/su17041486.
[28] B. Zongo, B. Barbier, A. Diarra, M. Zorom, C. Atewamba, O.S. Combary, S. Ouédraogo, P. Toé, Y. Hamma, T. Dogot, Economic analysis and food security contribution of supplemental irrigation and farm ponds: evidence from northern Burkina Faso, Agric. Food Secur. 11 (2022), https://doi.org/10.1186/s40066-021-00347-0.
[29] M. Sanogo, R. Yonaba, A. Lawane, M. Zorom, F. Tassembédo, H. Ali Sahad, I. Bazié, Do runoff water harvesting ponds affect farmers cropping choices? Insights from smallholders in the west African Sahel, Sustainability 16 (2024), https://doi.org/10.3390/su16188000.
[30] S. Halder, S. Das, S. Basu, Effectiveness of hapa as small rainwater harvesting ponds for sustainable agricultural development in diversified arid regions: comparative case studies from Bankura in India, J. Rural. Dev. 42 (2023) 198–217, https://doi.org/10.25175/jrd/2023/v42/i3/173263.

[31] J.B. Nana, H.M. Abd El Baki, H. Fujimaki, Designing an economical water harvesting system using a tank with numerical simulation model WASH_2D, Agronomy 14 (2024), https://doi.org/10.3390/agronomy14112466.
[32] M. Manikandan, B.B. Saliha, B. Narsimlu, J.V.N.S. Prasad, K. Baskar, V. Sanjivkumar, S. Manoharan, G. Guru, G.R. Chary, K.V. Rao, et al., Design of rainwater harvesting pond for runoff storage and utilization in semi-arid vertisols, Water 17 (2025), https://doi.org/10.3390/w17213034.
[33] A. Deo, S. Karmakar, A. Arora, Rainwater harvesting and water balance simulation-optimization scheme to plan sustainable second crop in small rain-fed systems, J. Environ. Manag. 323 (2022), https://doi.org/10.1016/j.jenvman.2022.116135.
[34] M. Kang, C. Yoo, W. Na, Assessment of the contribution of rainwater collection to crop production on Udo island, Korea, Water 13 (2021), https://doi.org/10.3390/w13223299.
[35] N. Noor, Z. Wanchang, N.H. Chandio, G.A. Mahar, G. Rahman, Conservation and harvesting of rainwater for sustainable agriculture in Kachho, Sindh, Pakistan, Ecol. Quest. 36 (2025) 1–15, https://doi.org/10.12775/EQ.2025.032.
[36] P. Panwar, D. Machiwal, V. Kumari, S. Kumar, P. Dogra, S. Manivannan, P. R. Bhatnagar, J.M.S. Tomar, R. Kaushal, D. Jinger, et al., Sustainable water harvesting for improving food security and livelihoods of smallholders under different climatic conditions of India, Sustainability 15 (2023), https://doi.org/10.3390/su15129230.
[37] J. Koide, S. Yokoyama, S. Hirouchi, C. Hirose, N. Oka, M. Oda, S. Yanagihara, Exploring climate-resilient and risk-efficient cropping strategies using a new pond irrigation system: an experimental study in northern Ghana, Agric. Syst. 191 (2021), https://doi.org/10.1016/j.agsy.2021.103149.
[38] D.R. Palsaniya, S. Kumar, M.M. Das, S.K. Rai, T.K. Kumar, M. Chaudhary, K. Chand, A. Ahmed, C.S. Sahay, P. Kumar, Rain water harvesting, agroforestry and goat based intensification for livelihood resilience in drought prone rainfed smallholder farming system: a case for semi-arid tropics, Agrofor. Syst. 97 (2023) 1405–1419, https://doi.org/10.1007/s10457-023-00863-x.
[39] W. Su, B. Li, X. Ouyang, S. Wu, H. Feng, K.H.M. Siddique, Integrated effects of rainwater harvesting and organic fertilization on soil structure and aggregates in a rainfed orchards on the loess plateau, Agric. Water Manag. 323 (2026) 110088, https://doi.org/10.1016/j.agwat.2025.110088.
[40] D.R. Palsaniya, S. Kumar, M.M. Das, T.K. Kumar, M. Chaudhary, K. Chand, S.K. Rai, A. Ahmed, C.S. Sahay, Ecosystem services from rain water harvesting, agroforestry and livestock based smallholder rain-fed integrated farming system, Agrofor. Syst. 98 (2024) 2617–2632, https://doi.org/10.1007/s10457-024-01055-x.
[41] A. Das, D. Datta, T. Samajdar, R.G. Idapuganti, M. Islam, B.U. Choudhury, K. P. Mohapatra, J. Layek, S. Babu, G.S. Yadav, Livelihood security of small holder farmers in eastern Himalayas, India: pond based integrated farming system a sustainable approach, Curr. Res. Environ. Sustain. 3 (2021) 100076, https://doi.org/10.1016/j.crsust.2021.100076.
[42] S.K. Rautaray, Field Design for Enhancing Water Productivity in waterlogged areas with efficient water harvesting and farming system, Agric. Res. 10 (2021) 255–261, https://doi.org/10.1007/s40003-020-00493-0.
[43] D. Rizzo, F. Vinatier, F. Jacob, I. Ferchichi, I. Mekki, J. Albergel, J.-S. Bailly, A framework for the sustainable maintenance of permanent runoff management structures in rainfed agriculture under climate change, J. Environ. Manag. 377 (2025) 124718, https://doi.org/10.1016/j.jenvman.2025.124718.
[44] S.F. Demessie, M. Woldeyohannes, Evaluation of water productivity of smallholder irrigation along a Sand River, in the Maigobo watershed of northern Ethiopia, Environmental Challenges 14 (2024) 100801, https://doi.org/10.1016/j.envc.2023.100801.
[45] A. Mullo-Sinaluisa, C. Oquendo-Borbor, A. Velastegui-Montoya, B. Merchan-Sanmartín, M. Chávez-Moncayo, V. Herrera-Matamoros, P. Carrión-Mero, Hill dam design to improve water use in rural areas—case study: Sacachún, Santa Elena, Sustainability 14 (2022), https://doi.org/10.3390/su141912268.
[46] D.H. Ranade, M.L. Jadav, I. Swarup, O.P. Girothia, D.V. Bhagat, A. Upadhyay, Crop productivity enhancement under soybean based cropping system through harvested rain water in Malwa region, Legum. Res. 45 (2022) 1440–1444, https://doi.org/10.18805/A-5447.
[47] M.F. Jilito, D.Y. Wedajo, B.W. Feyisa, T.W. Tuke, Water storage practices for small-scale irrigation systems in east Hararghe zone, Ethiopia, Water Sci. Technol. Water Supply 21 (2021) 1674–1686, https://doi.org/10.2166/WS.2021.017.
[48] L. Rosa, L. He, Global multi-model projections of green water scarcity risks in rainfed agriculture under 1.5 °C and 3 °C warming, Agric. Water Manag. 314 (2025) 109519, https://doi.org/10.1016/j.agwat.2025.109519.
[49] S. Chen, Q. Li, H. Sun, X. Chen, Global trends and future directions in rainwater harvesting research: insights from a joint bibliometric and literature review, Environ. Dev. Sustain. (2026) 1–34.